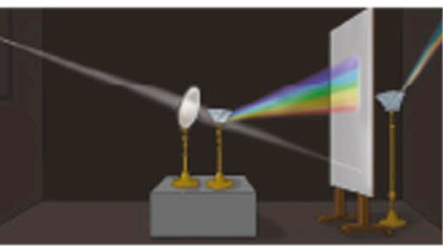


1670 ▶ 1690

1672

Rainbow colors

English physicist Isaac Newton (see pp.88–89) published a paper on light. He described an experiment he had carried out using two prisms to show that white light is made up of the seven colors of the rainbow.



Newton's sketch of his experiment with prisms

Calculating machines
See pages
124–125

1672

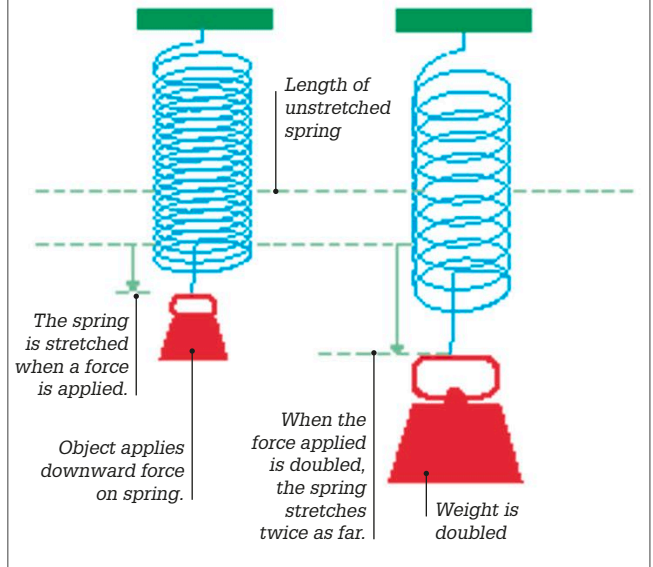
Calculating questions

Gottfried Leibniz, a German mathematician, created a calculating machine called the step reckoner. In 1674, he invented calculus, the mathematics of infinitesimal (very small) changes. Isaac Newton (see pp.88–89) also devised a version of calculus, and the two men fell out over who had done so first.

1678

Hooke's law

English scientist Robert Hooke observed that the force needed to stretch a spring is proportional to the distance it stretches. If the force is doubled, the distance is doubled. There is a point beyond which the spring does not stretch, but snaps.



1670

1675

1675

Astronomer to the king

England's King Charles II appointed John Flamsteed the first Astronomer Royal to head a new observatory in Greenwich, London. This observatory marked what would later become the Prime Meridian (0° longitude) between east and west.



Royal Observatory, Greenwich, UK

1679

Steam digester

French inventor Denis Papin demonstrated a cooking device that used high-pressure steam to extract fat from bones. A forerunner of the modern pressure cooker, Papin's digester was fitted with a steam release valve and a piston, leading in time to the development of the steam engine.

Lens for viewing
microscopic
samples

Screw adjusts
position of
specimen.



1676

Leeuwenhoek's
microscope

Life in miniature

Antoni van Leeuwenhoek, a Dutch merchant, made his own microscopes. He used them to observe tiny living creatures swimming in a drop of water, which he called "animalcules." These were, in fact, single-celled protozoans called amoeba.